Arrays

01 April 2026 16:35

- Primitive
  - int
  - boolean
  - char
  - float

- Non-primitive
  - array
  - String

[2, 3, 4, 5] → [2, 3, 4, 'a', true] ✗

Syntax

```
int [] arr = {1, 2, 3, 4};
```

- arr → name of array
- {1, 2, 3, 4} → [1, 2, 3, 4]

```
int n = ip();
int [] arr = {new int [n];
```

OS
- Stack memory
- Heap memory

Stack

a	5
	'a'

main

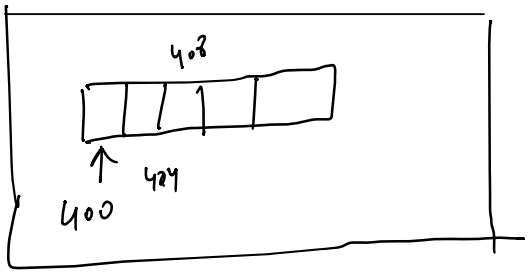
```
static void main() {
    int a = 5
```

Stack

a	5
b	'a'
flag	true
arr	400

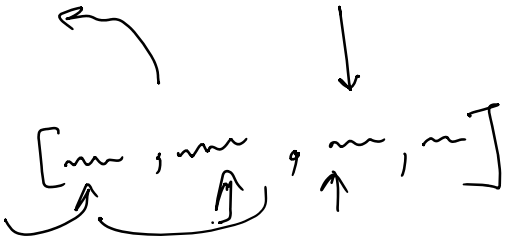
```
static void main(...)
    int a = 5
    char b = 'a';
    boolean flag = true;
    int [] arr = new int[5];
}
```

heap



int

Char → 2



arr = [1, 2, 3, 4, 5]

0 1 2 3 4

sout(arr[4])

arr[0]

size = 5

0 → 4

0
1
2
3
4

traversing an array:

arr[0]
arr[1]
arr[2]

i = 5

fn → 5

```
for (int i=0; i< arr.length; i++){
    sout( arr[i] );
```

0, 1, 2, 3, 4

[1, 2, 3, 4]

```
Sout( arr[i] );
}
```

[1, 2, 3, 4]

1
2
3
4

Q. Sum of the array.

[1, 2, 3, 4] → 10

```
int Sum = 0;
for(int i=0 ; i< arr.length ; i++) {
    Sum = Sum + arr[i];
}
Sout(sum);
```

[1, 2, 3, 4]

i=0
sum=0
Sum=1

i=1, Sum=1

1+2, Sum=3

i=2

i=2 → Sum=3+3=6

i=3, Sum=6

Sum= 6+4

Sum=10,

Sout(Sum);

Q. Highest number in an array.
print

[1, 2, 9, 6, 8]

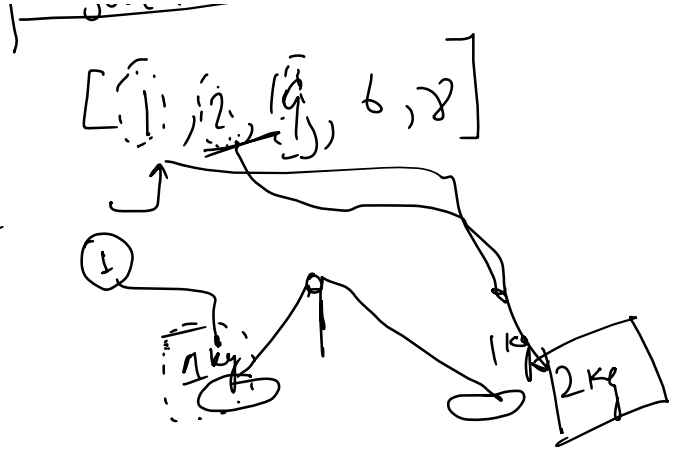
op → 9

index

[1, 2, 9, 6, 8]

```
int ans = arr[0];
for(int i = 0; i< arr.length ; i++){
    if ( arr[i] > ans) ans=arr[i];
}
```

if ([illegible])

}

Q. Print the index of the highest element.